

CMSC 207: Chapter 4 Homework

March 30, 2026

1. Disprove the following statement. Then write the negation of the following statement, and prove that the negation of the statement is true.

$\forall$ integers m and n , if $2m + n$ is odd then m and n are both odd.

Counterexample:

Choose integers $m = 0$ and $n = 1$. Then:

$$\begin{aligned} 2m + n &= 2(0) + 1 && \text{by substitution} \\ &= 1 && \text{by arithmetic of rhs} \end{aligned}$$

$m = 0$ is an even integer since it can be represented in the form $2k$ for some integer k .

1 is an odd integer since it can be represented in the form $2k + 1$ for some integer k .

Thus, we have found integers m and n such that $2m + n$ is odd but m is not odd. Therefore, by counterexample, the statement is false. ■

Negation

$\exists$ integers m and n such that $2m + n$ is odd and (m is even or n is even).

Proof:

Choose integers $m = 0$ and $n = 1$. Then $2m + n = 1$.

$m = 0$ is an even integer since it can be represented in the form $2k$ for some integer k .

1 is an odd integer since it can be represented in the form $2k + 1$ for some integer k .

Thus, we have found integers m and n such that $2m + n$ is odd and m is even. Therefore, the negation of the statement is true. ■

2. Prove that the following statement is false:

The product of any two irrational numbers is irrational.

Counterexample:

It is known that $\sqrt{2}$ is irrational. Consider the product of $\sqrt{2}$ with itself:

$$\sqrt{2}\sqrt{2} = 2.$$

Since 2 is a rational number, we have found two irrational numbers whose product is rational. Therefore, the statement is false. ■